

MLFF PERF-P2R CPU/Control-Plane Benchmark

2026-08-15

Scope

This benchmark measures the CPU/control-plane portion of PERF-P2R. It does **not** execute MACE training or inference and therefore does not close whole-funnel performance qualification.

The timed path is the authenticated DATA8 immutable fixed-file cache on the deterministic DATA8 regression fixture. The successive-fidelity work accounting follows the resource-allocation principle of successive halving.¹ The same DATA8 authority is rebuilt at the same output path for every sample. Fifteen fresh builds are compared with fifteen cache-hit builds after one cache-population pass.

DATA8 fixed-file cache

Metric	Fresh construction	Authenticated cache hit
Samples	15	15
Median wall time	79.696 ms	17.333 ms
Observed range	76.212–84.553 ms	16.105–20.481 ms
Median process CPU	61.923 ms	17.325 ms
Median write characters	204,478	80,429
Median read characters	641,416	157,667

The median cache-hit wall time is **4.598x faster**, a **78.25% reduction** on this bounded fixture. Cache population took 105.047 ms. The cache contained 11 authenticated generations, 33 files, and 152,891 bytes.

¹Kevin Jamieson and Amee Talwalkar, “Non-stochastic Best Arm Identification and Hyperparameter Optimization,” *Proceedings of Machine Learning Research* 51, 240–248 (2016), <https://proceedings.mlr.press/v51/jamieson16.html>.

Every cache-hit `Data8PreparationBundle.to_dict()` was exactly equal to the fresh reference authority. The result therefore qualifies the implementation hot path, not a new scientific representation.

Successive-fidelity exposure

For admitted sizes A , coarse survivors S_4 , short finalists S_2 , and coarse boundary e_0 ,

$$W = e_0 \sum_{i \in A} K_i + (10 - e_0) \sum_{i \in S_4} K_i + 20 \sum_{i \in S_2} K_i.$$

For all seven default sizes, the exact exposure envelope is:

Coarse boundary	Smallest-size survivors	Saved vs. exhaustive	Largest-size survivors	Saved vs. exhaustive
3 epochs	69,888	85.67%	402,048	17.56%
4 epochs	84,224	82.73%	402,944	17.38%
5 epochs	98,560	79.79%	403,840	17.19%

The benchmark record also enumerates ladder widths 3 through 7 for all three coarse boundaries. These values are exact structure-epoch work counts. They are **not wall-time predictions**.

Interpretation

PERF-P2R removes redundant fixed-file construction and makes training work incremental across stage boundaries. The measured CPU cache benefit is large on the development fixture, while the analytical exposure bound confirms that the corrected halving rule still provides meaningful work reduction even when the largest candidates survive.

No GPU, VRAM, utilization, MACE throughput, or resumed-versus-uninterrupted endpoint claim is made here. Those measurements remain consolidated in FINAL-GPU1.

Reproduction

```
python benchmarks/benchmark_mlff_perf_p2r.py \
  --repeats 15 \
  --output benchmarks/mlff_perf_p2r_cloud_cpu_2026-08-15.json
```

The JSON record is the machine-readable authority for the numbers above.